

Course Project: Predicting Public Passenger Vehicle Inspection Outcomes

Course: Nossim Mitkadmim Belemidat Mechona - Ariel University

Student: Ricardo Caster

GitHub link with code:

https://github.com/RicardoCaster/nossim_mitkadmim_belemidat_mechona

Section 1 – Introduction

Problem and Significance

Public passenger vehicles play a critical role in daily transportation and public safety. Ensuring that these vehicles meet inspection standards is essential to prevent mechanical failures, reduce accidents, and maintain public trust. However, not all vehicles carry the same level of risk. Being able to anticipate which inspections are more likely to fail can support better planning, prioritization, and risk management.

The objective of this project is to build a data-driven model that predicts whether a public passenger vehicle inspection will result in a **PASS** or **FAIL**, using only information available prior to the inspection. The focus is placed on identifying high-risk inspections while handling real-world challenges such as class imbalance and heterogeneous data types.

Beyond prediction, the project also aims to explore the data from multiple perspectives:

- 1. Develop a Predictive Model:** Evaluate different supervised learning algorithms to accurately identify vehicles at risk of failing inspection.
- 2. Handle Imbalanced Data:** Address the reality that most vehicles pass, ensuring the model doesn't simply ignore the minority "failure" class.
- 3. Identify Fleet Profiles:** Use unsupervised learning to group vehicles into risk clusters and detect statistical anomalies.
- 4. Ablation Testing:** Empirically determine if the timing of a scheduled inspection influences the mechanical outcome.

Section 2 – Dataset and Features

Dataset Description

Certain types of Public Passenger Vehicles licensed by the City of Chicago must be inspected as part of the license renewal or change of vehicle associated with the license. The dataset was obtained from a public governmental source (Source link: <https://catalog.data.gov/dataset/public-passenger-vehicle-inspection-schedule>) and contains records of public passenger vehicle inspections. Each record includes vehicle-related

attributes, organizational information, inspection details, scheduling information, and the final inspection result.

The original dataset consisted of **35,310 records**. After filtering and cleaning, **25,567 inspections** were retained for analysis.

Target Definition

Inspection outcomes include several categories (e.g., PASSED, FAILED, PENDING, NO_SHOW). For this project, only **PASSED** and **FAILED** inspections were retained, as they represent completed inspections with definitive outcomes. The target variable was encoded as a binary label:

- 0 → PASSED
- 1 → FAILED

FAILED inspections were treated as the positive class, since they represent risk and are operationally more critical.

Feature Selection and Rationale

Feature engineering was guided by domain reasoning rather than purely automated selection. The final feature set intentionally focused on **vehicle and organizational characteristics**, including:

- Vehicle type, make, model, and status
- Company and affiliation information
- Inspection type
- Vehicle age (engineered from model year)

Vehicle age was added as a meaningful mechanical proxy, as older vehicles are more likely to experience failures.

Time-based scheduling features (such as inspection hour or month) were deliberately excluded from the main model and later evaluated separately in an ablation study.

Preprocessing

- Missing numerical values were imputed using the median.
- Missing categorical values were handled using a dedicated "missing" category to preserve information.
- Categorical features were converted using one-hot encoding.
- Numerical features were standardized to ensure compatibility with scale-sensitive models.

All preprocessing steps were implemented within a unified pipeline to prevent data leakage.

Section 3 – Methodology

Supervised Learning Models

Multiple supervised learning algorithms were evaluated, each representing a different modeling philosophy:

- **Logistic Regression:** A simple and interpretable linear baseline.
- **Random Forest:** An ensemble of decision trees capable of modeling non-linear relationships.
- **XGBoost:** A gradient-boosted tree model known for strong performance on tabular data.
- **Support Vector Machine (SVM):** A margin-based classifier effective in high-dimensional spaces.
- **K-Nearest Neighbors (KNN):** A similarity-based model relying on local neighborhoods.

Strategic Choice: We utilized **Class Weight Balancing** for Logistic Regression, RF, and SVM. Since failures are the minority (~34%), standard models tend to be biased toward passes. Balancing penalizes the model more for missing a failure than for a false alarm.

Unsupervised Learning

Two unsupervised techniques were used to complement supervised predictions:

- **K-Means Clustering:** To identify latent risk profiles without using inspection outcomes.
- **Isolation Forest:** To detect rare or unusual inspections that may represent exceptional risk.

These approaches provide structural insight into the data beyond pure prediction.

Ablation Study

An ablation study was conducted to evaluate whether inspection scheduling time features (inspection hour and month) add predictive value. A simple Logistic Regression model and ROC-AUC metric were used to isolate the effect of these features without confounding factors.

Section 4 – Experiments and Results

Experimental Design

The dataset was split into training (80%) and testing (20%) sets using stratified sampling to preserve class proportions. All models were trained and evaluated using the same preprocessing pipeline to ensure fair comparison.

Evaluation Metrics

Given the imbalanced nature of the dataset, multiple metrics were used:

- **Accuracy:** Overall correctness (used cautiously).
- **Precision:** Reliability of predicted failures.
- **Recall:** Ability to detect actual failures (most critical).
- **F1-score:** Balance between precision and recall.
- **ROC-AUC:** Overall ranking quality across thresholds.
- **PR-AUC:** Performance on the minority class.

Preferred Metrics:

- **Recall:** Our primary metric. In safety, missing a failure (False Negative) is worse than a false alarm.
- **PR-AUC (Precision-Recall AUC):** More appropriate than ROC-AUC here because it focuses specifically on the model’s ability to handle the minority class (failures).

Supervised Model Performance

Model	Accuracy	Recall (Catching Failures)	ROC-AUC	PR-AUC
XGBoost	0.713	0.399	0.752	0.584
SVM	0.663	0.792	0.750	0.576
Logistic Regression	0.676	0.699	0.750	0.588

Comprehensive Findings

- **Algorithm Performance:** SVM was the most effective for safety goals, catching 79.2% of failures.
- **Logistic Regression** also achieved a high recall

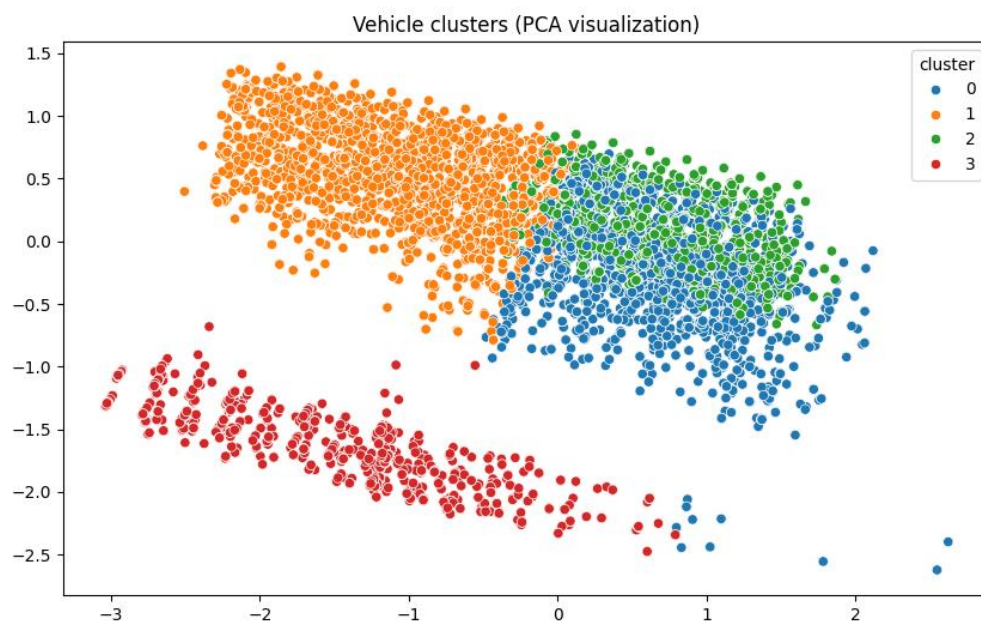
- **XGBoost** achieved the highest accuracy and ROC-AUC but at the cost of being too conservative(lower recall), missing 60% of failures.
- **Random Forest and KNN** underperformed, particularly in recall.
- **Ablation Study Result:** Adding inspection_month and inspection_hour only moved the ROC-AUC from **0.750 to 0.751**. This proves that a vehicle fails because of what it is (age/make), not *when* it is scheduled.

Unsupervised Results

Clustering revealed meaningful variation in failure rates across clusters, indicating distinct vehicle risk profiles.

To gain deeper insights into the underlying structure of the Chicago vehicle fleet, we applied **K-Means Clustering** followed by **Principal Component Analysis (PCA)** for dimensionality reduction. Because our feature set included high-dimensional one-hot encoded categorical variables, PCA was essential to project the data into a two-dimensional space for visual validation.

The PCA visualization revealed 4 distinct clusters:



The numbers on the axes (e.g., -3 to 3) represent **Standard Deviations** from the mean after data scaling:

- **The Center (0,0):** Represents the "Average Vehicle" in the fleet.

- **Coordinates:** Values like **-2** or **+2** show how many units a vehicle sits away from the average. A point at **-3** is a significant statistical outlier compared to the rest of the fleet.
- **Separation:** The distance between the **Red Cluster (3)** and the others mathematically proves that these vehicles have fundamentally different mechanical or organizational characteristics.

Average failure rate per cluster:

```
Average failure rate per cluster:
cluster
0    0.352554
1    0.319995
2    0.343115
3    0.371691
```

Cluster 3 showed the highest failure rate (**37.2%**), identifying it as a higher-risk profile.

The Isolation Forest detected 512 anomalies (approximately **2%** of inspections), including both PASSED and FAILED cases, suggesting the presence of rare but informative patterns.

Sample anomalies:

```
Sample anomalies:
Public Vehicle Type Vehicle Make vehicle_age Result
0           Ambulance      FORD          8 PASSED
1           Taxi          TOYOTA          9 PASSED
2           Taxi          NISSAN          7 PASSED
3           Taxi          FORD         16 PASSED
4           Taxi          TOYOTA          8 FAILED
```

Ablation Study Results

Including time-based scheduling features resulted in a ROC-AUC of **0.751**, nearly identical to the baseline model. This demonstrates that scheduling time does **not** meaningfully contribute to predicting inspection outcomes.

Section 5 – Conclusion and Discussion

Summary of Contributions

This project developed a complete end-to-end machine learning pipeline for predicting vehicle inspection outcomes. It combined supervised learning, unsupervised analysis, anomaly detection, and an ablation study to provide both predictive performance and interpretability.

Key findings include:

- Vehicle and organizational features are strong predictors of inspection outcomes.
- Recall-oriented models are preferable for safety-critical contexts.
- Scheduling time features do not add meaningful predictive value.
- Unsupervised methods provide valuable complementary insights.

Implications

The **SVM** model provides a viable path for the city to implement a "risk-based" inspection schedule. By prioritizing vehicles flagged by the model, the city could potentially identify nearly 80% of dangerous vehicles while only inspecting a fraction of the total fleet.

Team Contributions

This project was completed individually. All stages - including data preprocessing, modeling, evaluation, and analysis - were designed and implemented by the student. Online resources and large language models were used as supportive tools

Future Work

Future directions may include:

- Incorporating additional mechanical or historical inspection data, which was unavailable in this public dataset.
- Exploring Tabular Neural Networks (like TabNet) could potentially capture even more complex interactions between companies and vehicle models.
- Integrating model outputs into decision-support systems for inspection planning.
- A real-time API could be developed to flag high-risk vehicles the moment a company schedules an inspection online.